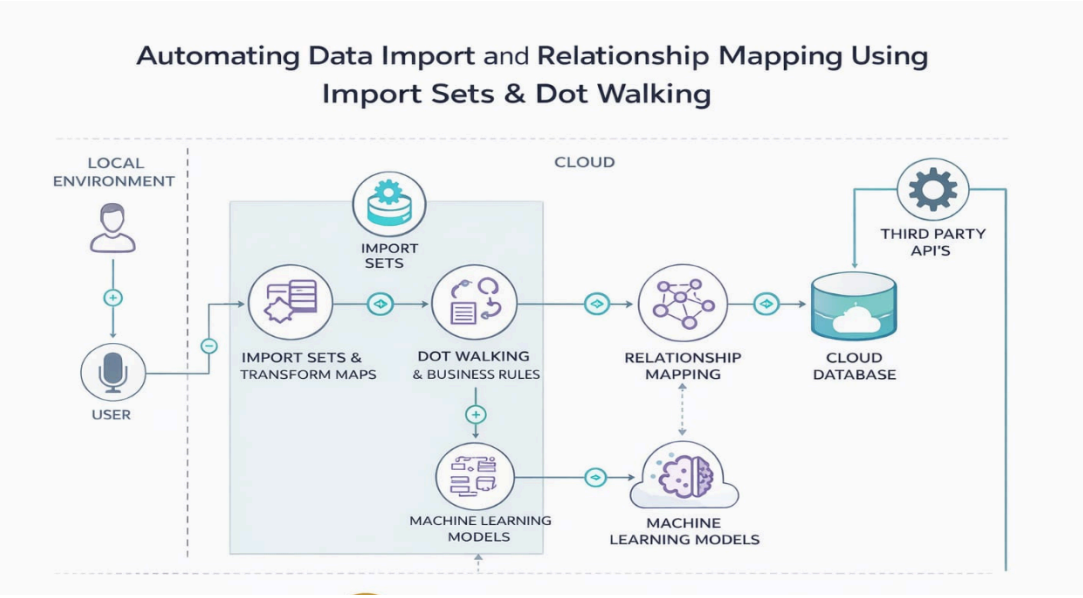
Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	24 March 2026
Team ID	69bf7f20c1eab70902258904
Project Name	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
Maximum Marks	4 Marks

Technical Architecture

The system automates data import into ServiceNow using Import Sets and Transform Maps. Uploaded data files are processed, transformed, and mapped into target tables. Dot walking is used to establish relationships between related tables.



S.No	Component	Description	Technology
1	User Interface	Interface for uploading data and viewing mapped relationships	HTML, CSS, JavaScript, ServiceNow UI
2	Application Logic-1	Import Set creation and data loading process	ServiceNow Import Sets
3	Application Logic-2	Transform Map logic for data transformation	ServiceNow Transform Maps
4	Application Logic-3	Relationship mapping using dot walking	ServiceNow Scripting (Glide Record, Dot Walking)
5	Database	Stores imported and transformed data	ServiceNow Tables
6	Cloud Database	Cloud-based data storage and management	ServiceNow Cloud Platform
7	File Storage	Stores uploaded files (CSV, Excel)	ServiceNow Attachment System
8	External API-1	Fetch external data if required	REST API Integration
9	External API-2	Third-party system integration	SOAP / REST APIs
10	Machine Learning Model	(Optional) Data validation or prediction	Not Applicable / Basic Rule Engine

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for development	ServiceNow Platform (Low-code), JavaScript
2	Security Implementations	Access control and data protection	Role-Based Access Control (RBAC), ACL, Encryption, OWASP Practices